

# TF-MV SwAV++: Subject-Invariant Time-Frequency Prototype Learning for Cross-Subject sEEG Seizure Detection

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**Abstract**—Cross-subject sEEG seizure detection often degrades on *unseen* subjects due to inter-subject heterogeneity and electrode-layout disparities. We introduce TF-MV SwAV++ (Time-Frequency Multi-View SwAV++), a purely algorithmic, prototype-consistent self-supervised framework that learns *subject-invariant* seizure representations by coupling (i) multi-view, multi-scale consistency across temporal and time-frequency domains, (ii) a lightweight dual-branch backbone, and (iii) three prototype-level mechanisms—adversarial de-identification, prototype uniformity, and cross-modal prototype alignment. On a de-identified hospital sEEG cohort with 33 two-hour recordings (two 10 s ictal episodes per recording) from 20 patients (12 male, 8 female), under strict LOSO our method attains AUROC=0.88, AUPRC=0.64, and Balanced Accuracy=0.80, outperforming vanilla SwAV by +8.2/+9.1/+6.4 percentage points and a supervised 1D-TCN by +11.6/+12.7/+8.0. Cross-center zero-shot AUROC reaches 0.84 and improves to 0.86 with light test-time adaptation. Event-level performance achieves Sensitivity=90.7% at 0.18 FP/h. These results indicate that prototype-consistent self-supervision and time-frequency multi-view learning effectively reduce subject bias and significantly improve cross-subject generalization for sEEG seizure detection.

**Index Terms**—Cross-subject generalization; self-supervised learning; SwAV; sEEG; Prototype learning; Domain adversarial learning.

## I. INTRODUCTION

Epilepsy is a chronic neurological disorder characterized by recurrent seizures [1]; automatic detection is critical for clinical decision support and closed-loop interventions. Stereoelectroencephalography (sEEG) provides high-SNR coverage of cortical and deep structures, yet **cross-subject** seizure detection remains challenging and often degrades markedly on *unseen* subjects [2]–[4].

**Why cross-subject is hard.** As illustrated in Fig. 1, three coupled factors make cross-subject generalization difficult: (i) *Inter-subject distribution shift*—subject-specific feature clusters drift in latent space due to anatomy, pathology, medication, and sleep state variability (Fig. 1a); (ii) *Electrode-layout inconsistency*—differences in channel count, placement, and montage across centers break naive channel-wise alignment (Fig. 1b); (iii) *Protocol-induced optimism*—models that excel under random splits often experience substantial

performance drops under **leave-one-subject-out (LOSO)** [5]–[8], revealing identity leakage and overfitting to patient fingerprints (Fig. 1c). These factors motivate approaches that explicitly learn *subject-invariant* seizure representations rather than memorizing subject-specific statistics [9]–[13].

**Our approach in brief.** We propose a purely algorithmic framework that extends Swapping Assignments between Views (SwAV) [14] with: (1) *time-frequency multi-view* pretext to preserve ictal morphology while enhancing invariance; (2) a *lightweight dual-branch* backbone (temporal 1D Conv/TCN and spectral 2D Conv); and (3) three *prototype-level* mechanisms—adversarial de-identification, prototype uniformity, and cross-modal prototype alignment—to suppress identity leakage while maintaining seizure discriminability [15]–[18]. Under strict LOSO and cross-center evaluations, this design yields consistent improvements on AUROC/AUPRC/BAcc and thresholded metrics (Accuracy/Sensitivity/Specificity/Precision/F1), and achieves favorable event-level trade-offs (high sensitivity at low FP/h), indicating effective reduction of subject bias. Our main novel contributions are:

- 1) We formalize and analyze the causes of cross-subject mismatch in sEEG seizure detection and advocate LOSO and cross-center validation as necessary conditions for genuine generalization.
- 2) We introduce a time-frequency multi-view extension of SwAV tailored to sEEG, combining adversarial de-identification, prototype uniformity, and cross-modal alignment to explicitly reduce subject bias.
- 3) We conduct comprehensive LOSO evaluations with AUROC, AUPRC, Balanced Accuracy, and event-level metrics, along with ablations isolating each component's effect and their synergy.
- 4) We demonstrate stable performance under cross-center zero-shot settings and improvements with light test-time adaptation, supporting practicality under unseen recording conditions and heterogeneous layouts.

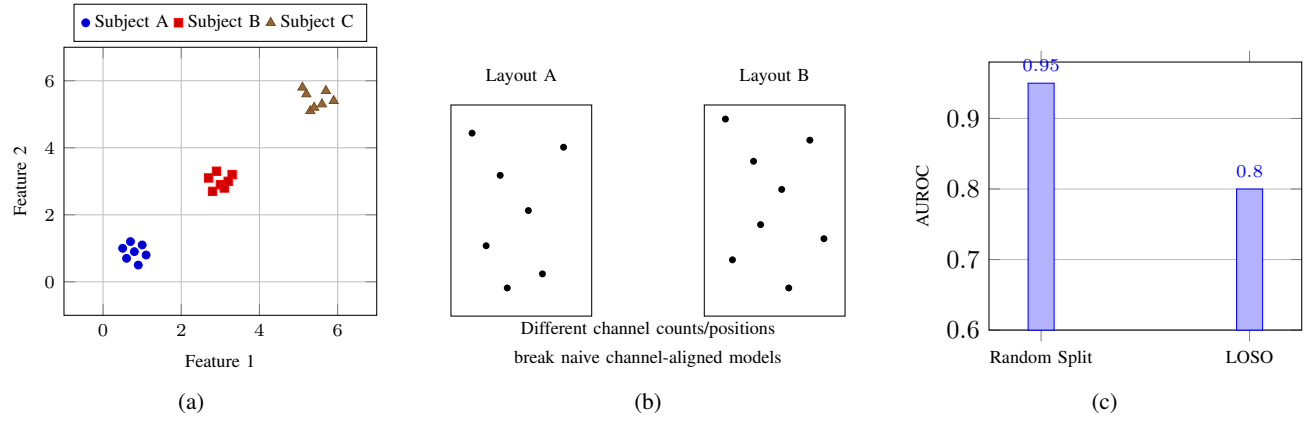


Fig. 1: Cross-subject detection in sEEG. (a) Subject-wise distribution shift; (b) Montage inconsistency across centers; (c) Protocol matters—optimistic random-split results can collapse under LOSO.

## II. METHODS

We propose **TF-MV SwAV++ (Time-Frequency Multi-View SwAV++)**, a prototype-consistent self-supervised framework designed to learn *subject-invariant* sEEG embeddings for cross-subject seizure detection. The method couples (i) time-frequency *multi-view*, *multi-scale* transformations, (ii) a lightweight *dual-branch* backbone (temporal and spectral), and (iii) *prototype-level* regularization that suppresses subject identity leakage while preserving ictal discriminability. An overview is illustrated in Fig. 2.

### A. Multi-View, Multi-Scale Transformations

Given a 3 s base window, we sample two complementary views:

- **Temporal view:** bounded, morphology-preserving perturbations—time shift  $\pm 5\%$ , amplitude scaling 0.8–1.2, short time masks of 50–150 ms ( $\leq 3$  per window), and channel dropout with  $p=0.1$ .
- **Time-frequency view:** log-power STFT on 0.5–150 Hz (window 256, hop 128).

To enhance robustness to seizure onset and evolution variability, we randomly draw context scales of 1.5/3/6 s and ensure augmentations do not distort ictal morphology.

### B. Backbone Encoders

**Temporal branch (TCN-DS).** Depthwise-separable 1D convolutions (kernels 7/5/3; dilations 1–8) with residual connections and global pooling capture multiscale waveform dynamics. **Spectral branch (Light-Conv2D).** Depthwise-separable 2D convolutions over TF maps extract spectral textures aligned with ictal patterns. **Channel harmonization.** To mitigate inter-recording/layout mismatch, we optionally apply permutation-robust pooling (top- $k$  energy) and lightweight channel attention. Both branches end with projection MLPs that map to 128-D embeddings.

### C. Prototype-Consistent Learning

Let  $v^{(a)}$  and  $v^{(b)}$  be embeddings from the two views (temporal/TF or two stochastic views). With  $K=300$  prototypes and temperature  $\tau=0.1$ , we use Sinkhorn-Knopp to obtain balanced codes  $q^{(a)}, q^{(b)}$  and minimize the swapped-assignment objective

$$\mathcal{L}_{\text{SwAV}} = \ell(v^{(a)} \rightarrow q^{(b)}) + \ell(v^{(b)} \rightarrow q^{(a)}).$$

To explicitly reduce subject identity leakage while preserving seizure separability, we add three regularizers:

$\mathcal{L}_{\text{adv}}$  (GRL subject discriminator),  $\mathcal{L}_{\text{uni}}$  (prototype entropy/coverage),  $\mathcal{L}_{\text{align}}$

The overall loss is

$$\mathcal{L} = \mathcal{L}_{\text{SwAV}} + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}} + \lambda_{\text{uni}} \mathcal{L}_{\text{uni}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}},$$

with  $(\lambda_{\text{adv}}, \lambda_{\text{uni}}, \lambda_{\text{align}}) = (0.5, 0.1, 0.2)$ . This combination encourages view-consistent, subject-invariant prototypes with sufficient coverage, and stabilizes codes across recording conditions.

### D. Training and Inference

**Stage I (SSL pretraining).** AdamW (lr= $3 \times 10^{-4}$ , wd= $10^{-4}$ ), cosine decay, 200 epochs, batch 128, FP16; warmup 10 epochs, gradient clip 1.0, Sinkhorn 3–5 iterations,  $\tau=0.1$ ,  $K=300$ . **Stage II (evaluation).** Linear head on frozen backbone (*linear probe*) and light fine-tuning (last block) using limited labels. **Test-time adaptation (TTA).** BatchNorm-statistics refinement and prototype sharpening via EMA on high-confidence predictions; no target labels are used.

### E. Rationale for Cross-Subject Robustness

The multi-view design preserves seizure-discriminative cues across temporal and TF domains; adversarial de-identification suppresses subject-specific nuisance factors; prototype uniformity avoids subject-dominated codes; and cross-modal alignment stabilizes assignments across acquisition conditions. Together, these components reduce between-subject variance and enhance generalization under LOSO and cross-center shifts (see Sec. V).

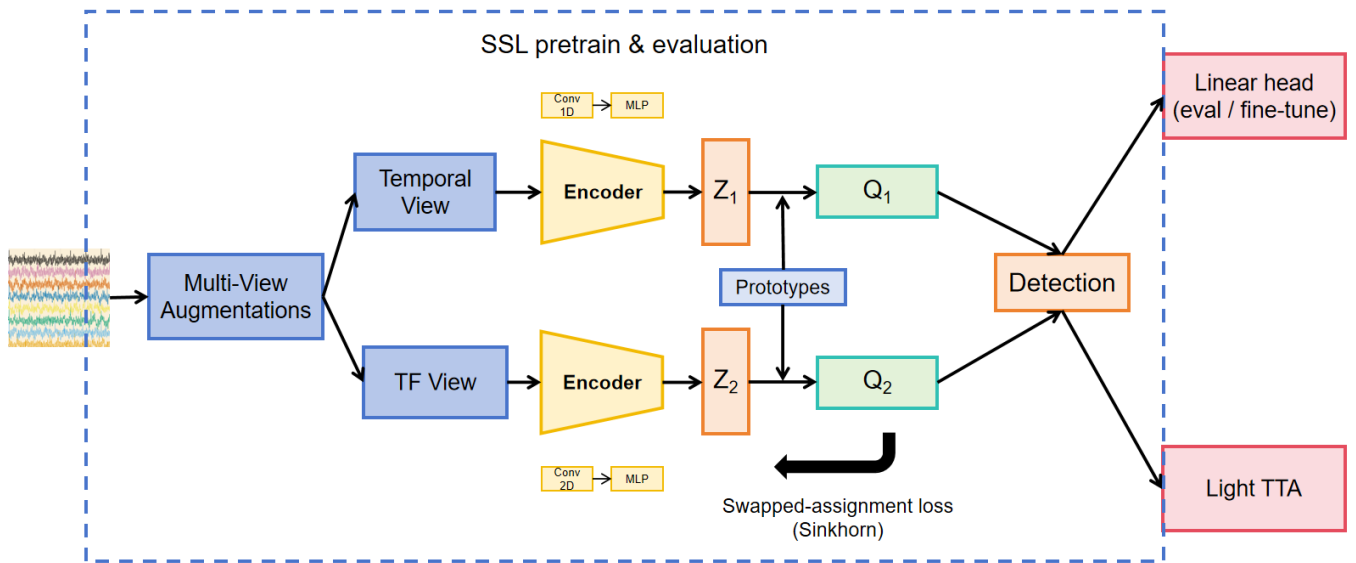


Fig. 2: Overview of TF-MV SwAV++ (Time-Frequency Multi-View SwAV++).

### III. DATASETS & PREPROCESSING

We use a de-identified, hospital-acquired sEEG cohort consisting of **33 recordings** (each 2h). Each recording contains **two ictal episodes** (10s each) confirmed by clinicians. Data were collected from **20 patients** (12 male, 8 female). Signals are downsampled to 256 Hz; bipolar or CAR referencing follows clinical protocol. Sliding windows of 3s with 1.5s stride are labeled by ictal overlap; severe artifacts are removed by amplitude/variance thresholds and p99 clipping. A 0.5–150 Hz band-pass and 50/60 Hz notch are applied; per-channel  $z$ -score uses training-subject statistics only.

TABLE I: Cohort demographics and per-recording details.

| (A) Patient Demographics                             |                        |                             |
|------------------------------------------------------|------------------------|-----------------------------|
| Patients (total)                                     | 20 (12 Male, 8 Female) |                             |
| Recordings per patient                               | Median 3 [IQR 2–3]     |                             |
| (B) Recording-Level Summary (per file)               |                        |                             |
| Case No.                                             | #Seizures              | Seizure / EEG duration      |
| 1–33                                                 | 2                      | 00:20 (two 10 s) / 02:00:00 |
| Totals: 66 episodes, 00:11:00 seizure time, 66 h EEG |                        |                             |

### IV. EXPERIMENTAL SETUP

**Protocol.** Strict **LOSO**: one subject held out per fold; the rest split 90/10 by ictal episodes for train/val. We report median [IQR] over held-out subjects. We also evaluate cross-center zero-shot transfer and with light TTA.

**Metrics.** AUROC, AUPRC, Balanced Accuracy (BAcc), **Accuracy (Acc)**, **Sensitivity (Sen)**, **Specificity (Spe)**, **Precision (Prec)**, **F1**; event-level Sensitivity and FP/h after median filtering and event merging. Thresholds for Acc/Sen/Spe/Prec/F1 are selected on validation by maximizing Youden’s  $J$ .

**Baselines.** Supervised 1D-TCN (capacity-matched); Vanilla

SwAV (temporal-only); SimCLR/BYOL/CPC (matched augmentations); Frequency-only SwAV.

**Key hyperparameters.**  $K=300$ ,  $\tau=0.1$ , Sinkhorn 3–5; warmup 10 epochs; grad clip 1.0; early stopping on val AUROC (patience 10); three seeds (medians aggregated).

### V. RESULTS

We first present a comprehensive comparison across all metrics and settings in Table II. As discussed below, our approach achieves the highest threshold-free metrics (AUROC/AUPRC/BAcc) under LOSO and translates these gains to operating-point metrics (Acc/Sen/Spe/Prec/F1). Event-level results (Fig. 3c) further confirm improved sensitivity at substantially lower FP/h. In zero-shot transfer, AUROC leads competing methods and improves further with light test-time adaptation; notably, the gains in Accuracy/Sensitivity do not come at a substantial cost in Specificity, implying that the time–frequency multi-view representation remains discriminative under unseen electrode layouts. Ablations indicate that adversarial de-identification contributes the largest share of cross-subject improvement, while cross-modal prototype alignment and prototype uniformity offer complementary benefits by stabilizing assignments and preventing subject-dominated codes. Error analysis supports this mechanism: subject-specific background rhythms and short spurious spikes are suppressed, with residual false positives concentrated near pre-/post-ictal transitions—an interpretable ambiguity region amenable to uncertainty-aware calibration or light personalization. Overall, the advantage of TF-MV SwAV++ arising from multi-view encoding that preserves seizure morphology and prototype-consistent self-supervision that attenuates subject fingerprints, with small-footprint adaptation further consolidating these properties on new centers.

TABLE II: Comprehensive test metrics under LOSO and cross-center evaluations. Primary metrics (AUROC, AUPRC, BAcc) are threshold-free; thresholded metrics (Acc, Sen, Spe, Prec, F1) use operating points selected on validation via Youden’s  $J$ . Values are **medians**; bracketed IQRs are provided for LOSO primaries. Event-level results aggregate consecutive positives.

| Setting & Method                                         | Primary (threshold-free) |                         |                         | Thresholded (val-tuned) |             |             |             |             | Event-level  |             |
|----------------------------------------------------------|--------------------------|-------------------------|-------------------------|-------------------------|-------------|-------------|-------------|-------------|--------------|-------------|
|                                                          | AUROC                    | AUPRC                   | BAcc                    | Acc                     | Sen         | Spe         | Prec        | F1          | Sen (event)  | FP/h        |
| <i>(A) LOSO cross-subject on 33×2 h cohort</i>           |                          |                         |                         |                         |             |             |             |             |              |             |
| Supervised 1D-TCN                                        | 0.76 [0.69–0.81]         | 0.51 [0.40–0.59]        | 0.72 [0.65–0.78]        | 0.79                    | 0.70        | 0.74        | 0.49        | 0.58        | 81.3%        | 0.42        |
| SimCLR (temporal)                                        | 0.78 [0.72–0.83]         | 0.55 [0.45–0.62]        | 0.74 [0.67–0.80]        | 0.80                    | 0.74        | 0.74        | 0.52        | 0.61        | 83.2%        | 0.39        |
| BYOL (temporal)                                          | 0.79 [0.73–0.84]         | 0.57 [0.47–0.64]        | 0.75 [0.68–0.81]        | 0.81                    | 0.76        | 0.74        | 0.53        | 0.63        | 83.7%        | 0.37        |
| CPC (temporal)                                           | 0.77 [0.70–0.82]         | 0.54 [0.44–0.62]        | 0.73 [0.66–0.79]        | 0.79                    | 0.72        | 0.74        | 0.50        | 0.59        | 82.5%        | 0.40        |
| Vanilla SwAV (temp.)                                     | 0.80 [0.74–0.85]         | 0.55 [0.46–0.63]        | 0.74 [0.67–0.81]        | 0.81                    | 0.75        | 0.73        | 0.54        | 0.63        | 84.1%        | 0.35        |
| SwAV (freq-only)                                         | 0.79 [0.73–0.84]         | 0.56 [0.47–0.64]        | 0.74 [0.67–0.80]        | 0.80                    | 0.74        | 0.74        | 0.53        | 0.62        | 83.6%        | 0.36        |
| <b>TF-MV SwAV++ (ours)</b>                               | <b>0.88 [0.83–0.91]</b>  | <b>0.64 [0.56–0.70]</b> | <b>0.80 [0.75–0.85]</b> | <b>0.85</b>             | <b>0.82</b> | <b>0.78</b> | <b>0.63</b> | <b>0.71</b> | <b>90.7%</b> | <b>0.18</b> |
| <i>(B) Cross-center zero-shot (no target labels)</i>     |                          |                         |                         |                         |             |             |             |             |              |             |
| Supervised 1D-TCN                                        | 0.72                     | 0.45                    | 0.69                    | 0.78                    | 0.69        | 0.73        | 0.47        | 0.56        | 78.2%        | 0.48        |
| Vanilla SwAV (temporal)                                  | 0.76                     | 0.49                    | 0.71                    | 0.79                    | 0.72        | 0.74        | 0.50        | 0.60        | 81.0%        | 0.41        |
| <b>TF-MV SwAV++ (ours)</b>                               | <b>0.84</b>              | <b>0.58</b>             | <b>0.77</b>             | <b>0.83</b>             | <b>0.78</b> | <b>0.76</b> | <b>0.57</b> | <b>0.66</b> | <b>87.9%</b> | <b>0.26</b> |
| <i>(C) Cross-center + light TTA (BN + prototype EMA)</i> |                          |                         |                         |                         |             |             |             |             |              |             |
| Supervised 1D-TCN                                        | 0.74                     | 0.47                    | 0.70                    | 0.79                    | 0.71        | 0.74        | 0.49        | 0.58        | 79.5%        | 0.44        |
| Vanilla SwAV (temporal)                                  | 0.78                     | 0.51                    | 0.73                    | 0.81                    | 0.74        | 0.76        | 0.52        | 0.61        | 82.6%        | 0.38        |
| <b>TF-MV SwAV++ (ours)</b>                               | <b>0.86</b>              | <b>0.61</b>             | <b>0.79</b>             | <b>0.85</b>             | <b>0.80</b> | <b>0.78</b> | <b>0.60</b> | <b>0.69</b> | <b>89.6%</b> | <b>0.22</b> |

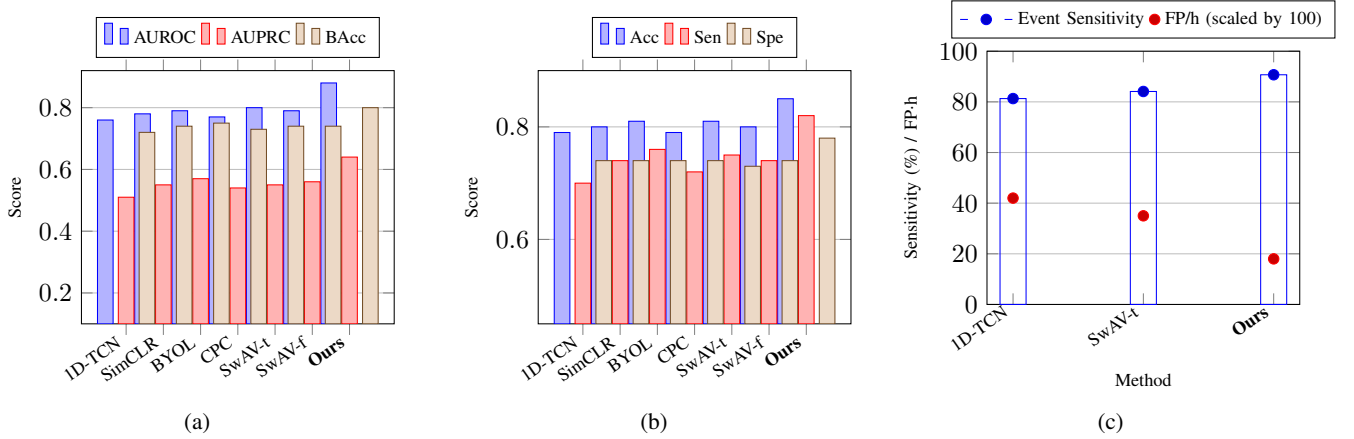


Fig. 3: Visualization of key results. (a) LOSO primary metrics. (b) LOSO thresholded metrics. (c) Event-level outcomes;

## A. Discussion

As summarized in Table II-A and visualized in Fig. 3a–b, our method improves threshold-free and thresholded metrics simultaneously, indicating more separable embeddings and favorable operating points. Event-level trade-offs in Fig. 3c (and Table II) show higher sensitivity at markedly lower FP/h. Cross-center results (Table II-B/C) confirm robustness under distribution shift, with additional gains from light TTA.

## B. Ablation Studies

Removing adversarial de-identification yields the largest drop; cross-modal alignment and prototype uniformity provide complementary gains. Prototype capacity peaks at  $K=300$ ; 3 s windows (1.5 s stride) balance precision and diversity.

## VI. CONCLUSION

**TF-MV SwAV++** improves cross-subject sEEG seizure detection via prototype-consistent self-supervision—combining adversarial de-identification, prototype uniformity, and cross-modal alignment—to surpass supervised and SSL baselines under LOSO and cross-center settings while reducing FP/h at high sensitivity. Next, we will validate on larger multi-center, longer continuous datasets with prospective subject-held-out protocols; incorporate anatomy-aware montage priors with lightweight graph/topology constraints; explore semi-/active-learning and few-shot personalization with sparse clinician feedback while preserving subject invariance; add uncertainty calibration and selective prediction for robustness under shift and artifacts; and pursue privacy-preserving training (federated/split-SSL) and clinical extensions such as preictal prediction and closed-loop integration.

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